# Term 3 Practice Activity: C5 Standard Normal Table

## Contents



## Evaluated criteria

- C5. Uses the normal standard distribution and its tables to calculate probabilities in contextual situations.

**Meaning of the notation.** The expression $P(Z < x)$ means the probability that the standard normal random variable $Z$ is less than the value $x$. In the graph of the standard normal distribution, this corresponds to the area under the curve to the left of $x$.

For example $P(Z < 0.85)$ means the area under the standard normal curve to the left of $x = 0.85$. Since the table gives values of $\Phi(z) = P(Z < x)$, this value can be read directly from the table as

$$P(Z < 0.85) = \Phi(0.85) = 0.8023.$$

Here are five examples of left-tail probabilities read directly from the table.

| $x$ | $P(Z < x) = \Phi(x)$ | Value |
|---|---|---|
| 0.25 | $P(Z < 0.25) = \Phi(0.25)$ | 0.5987 |
| 0.70 | $P(Z < 0.70) = \Phi(0.70)$ | 0.7580 |
| 0.85 | $P(Z < 0.85) = \Phi(0.85)$ | 0.8023 |
| 1.20 | $P(Z < 1.20) = \Phi(1.20)$ | 0.8849 |
| 1.65 | $P(Z < 1.65) = \Phi(1.65)$ | 0.9505 |

If the value is negative, such as $P(Z < -1.20)$, the expression means the area to the left of $z = -1.20$. Since the table only shows positive $z$-values, symmetry and the complement rule can be used:

$$P(Z < -1.20) = P(Z > 1.20) = 1 - P(Z < 1.20) = 1 - \Phi(1.20).$$

Since $\Phi(1.20) = 0.8849$,

$$P(Z < -1.20) = 1 - 0.8849 = 0.1151.$$

The same idea can be summarized as follows.

| $X$ | $-X$ | $P(Z < -X)$ | $1 - P(Z < -X)$ | $P(Z < X)$ |
|---|---|---|---|---|
| $-0.25$ | 0.25 | 0.5987 | 0.4013 | 0.4013 |
| $-0.70$ | 0.70 | 0.7580 | 0.2420 | 0.2420 |
| $-0.85$ | 0.85 | 0.8023 | 0.1977 | 0.1977 |
| $-1.20$ | 1.20 | 0.8849 | 0.1151 | 0.1151 |
| $-1.65$ | 1.65 | 0.9505 | 0.0495 | 0.0495 |

**Standard normal table (values of $\Phi(z)$).** Use this same table for all problems in this section.

| $z$ | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| 1.7 | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| 1.8 | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |
| 1.9 | 0.9713 | 0.9719 | 0.9726 | 0.9732 | 0.9738 | 0.9744 | 0.9750 | 0.9756 | 0.9761 | 0.9767 |
| 2.0 | 0.9772 | 0.9778 | 0.9783 | 0.9788 | 0.9793 | 0.9798 | 0.9803 | 0.9808 | 0.9812 | 0.9817 |

# 1 Problem 1: Left Tail with Positive and Negative Bounds

**Problem.** Let $Z \sim N(0,1)$.

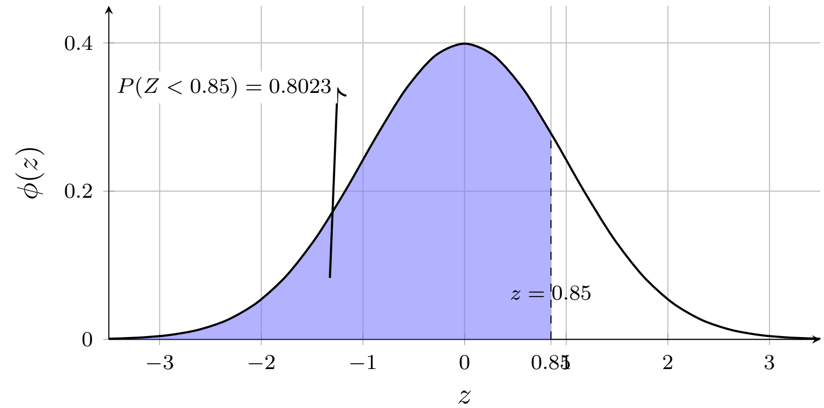
**Questions/Tasks.**

- **a)** Find $P(Z < 0.85)$ using the table.
- **b)** Find $P(Z < -1.20)$ using the table.

## Solution

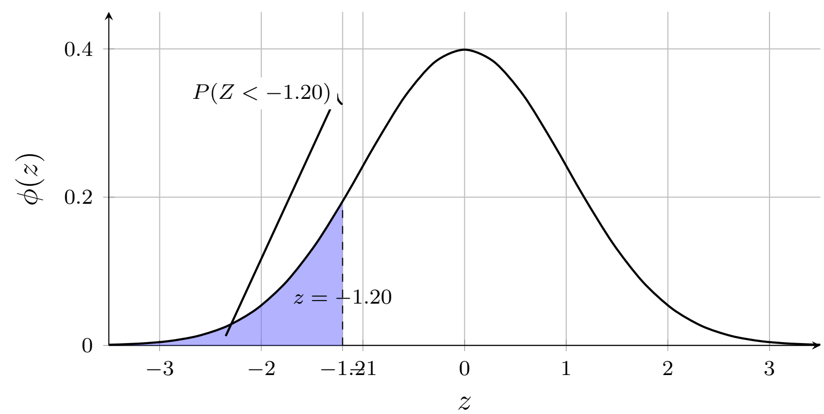
**a)** From the table, $\Phi(0.85) = 0.8023$, so

$$P(Z < 0.85) = 0.8023.$$

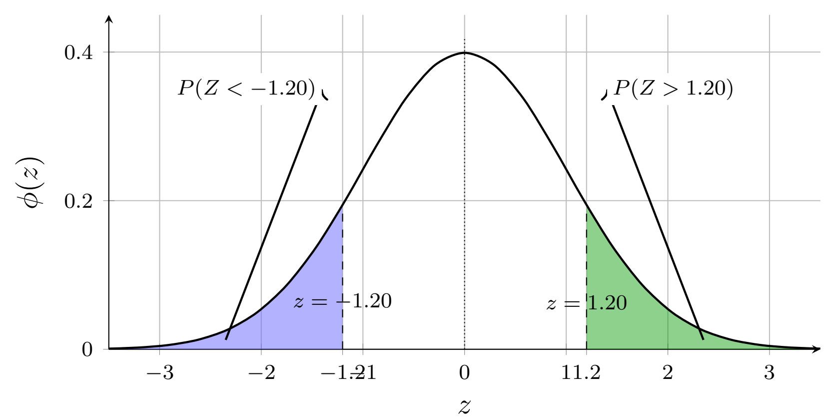



**b)** The goal is to find the left-tail probability

$$P(Z < -1.20).$$




By Symmetry: $P(Z < -1.20) = P(Z > 1.20)$.

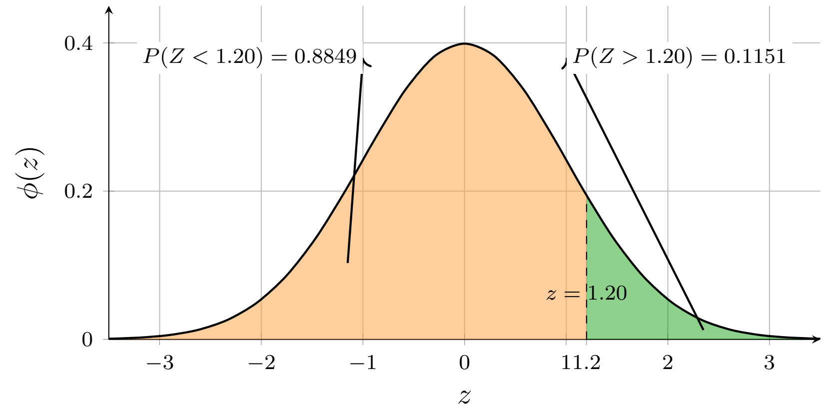



Now use the table value

$$P(Z < 1.20) = \Phi(1.20) = 0.8849.$$

Using the complement rule

$$P(Z > 1.20) = 1 - \Phi(1.20) = 1 - 0.8849 = 0.1151.$$




Therefore,

$$P(Z < -1.20) = P(Z > 1.20) = 0.1151.$$

**Interpretation.** About 80.2% of values are below 0.85, while 11.5% are below −1.20.

## 2 Problem 2: Right Tail with Positive and Negative Bounds

**Problem.** Let $Z \sim N(0, 1)$.

**Questions/Tasks.**

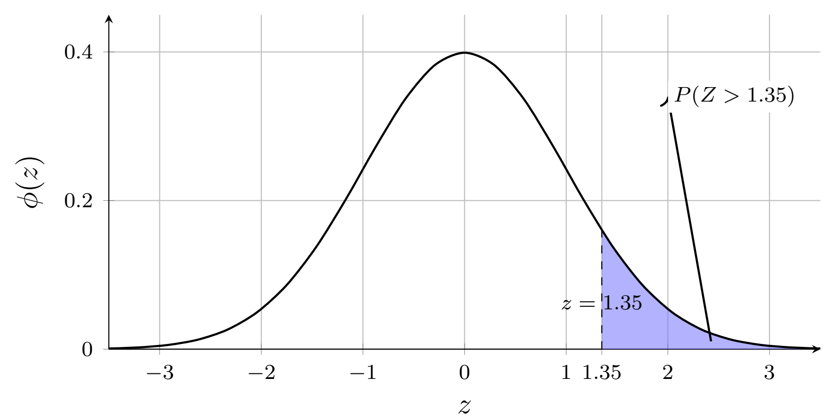
**a)** Find $P(Z > 1.35)$ using the table.

**b)** Find $P(Z > -0.70)$ using the table.

### Solution

**a)** The goal is to find the right-tail probability

$$P(Z > 1.35).$$

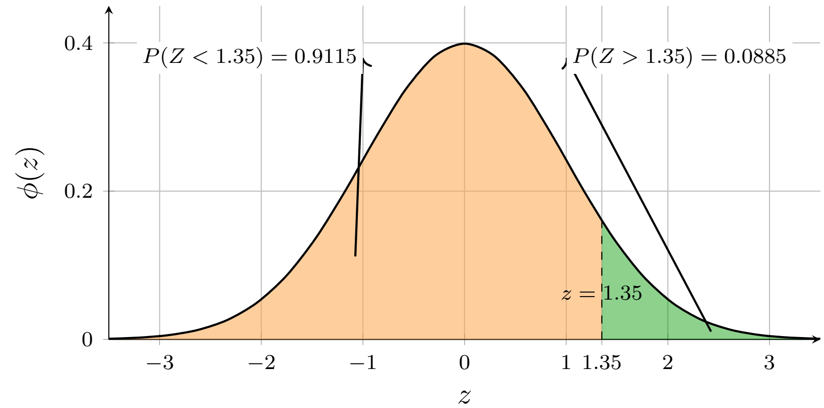



Now use the table value

$$P(Z < 1.35) = \Phi(1.35) = 0.9115.$$

Using the complement rule,

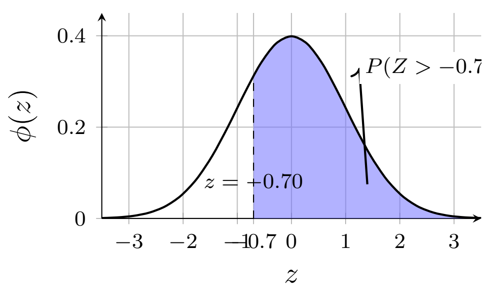
$$P(Z > 1.35) = 1 - \Phi(1.35) = 1 - 0.9115 = 0.0885.$$

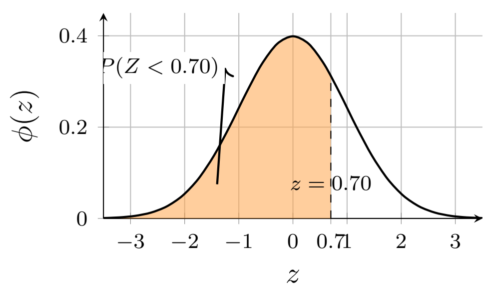



**b)** The goal is to find the right-tail probability

$$P(Z > -0.70).$$

The next two graphs show the wanted area and its mirrored left-tail image.




By symmetry of the standard normal distribution,

$$P(Z > -0.70) = P(Z < 0.70) = \Phi(0.70) = 0.7580.$$

**Interpretation.** The right tail above 1.35 is small (8.9%), but above -0.70 it is much larger (75.8%).

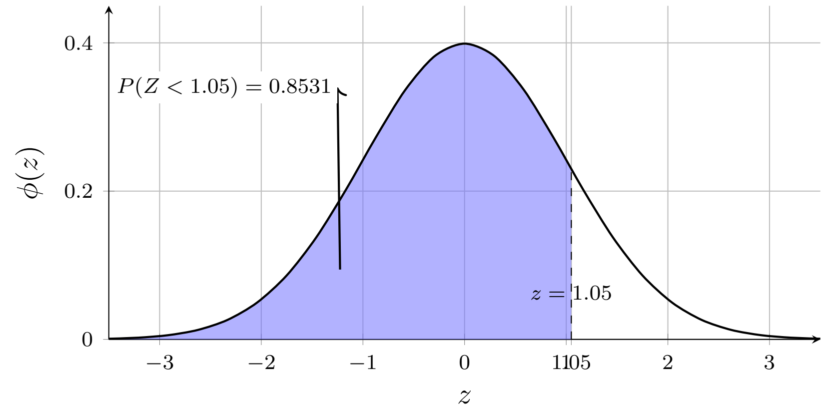
## 3 Problem 3: Mixed Left-Tail and Right-Tail Reasoning

**Problem.** Let $Z \sim N(0, 1)$.

**Questions/Tasks.** Calculate $P(Z < 1.05)$ and $P(Z > -0.55)$, and identify which is left-tail reasoning and which is right-tail reasoning.

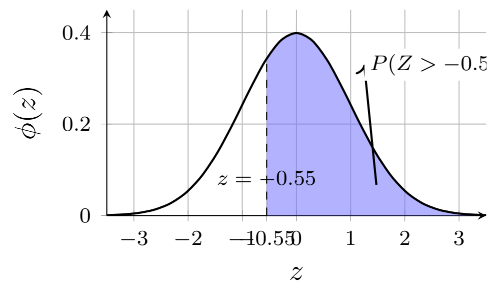
### C5

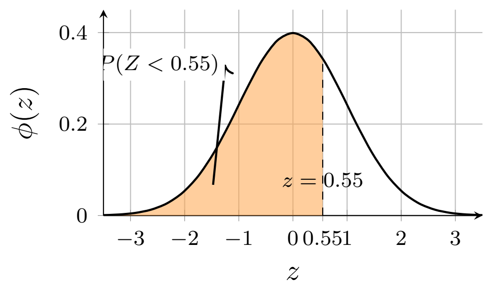
For $P(Z < 1.05)$, this is a left-tail probability:




$$P(Z < 1.05) = \Phi(1.05) = 0.8531.$$

For $P(Z > -0.55)$, this is a right-tail probability with a negative bound:




$$P(Z > -0.55) = P(Z < 0.55) = \Phi(0.55) = 0.7088.$$

**Interpretation.** Left-tail and right-tail setups use different formulas even when both answers are fairly large.

# 4 Problem 4: Central Band with Positive Bounds

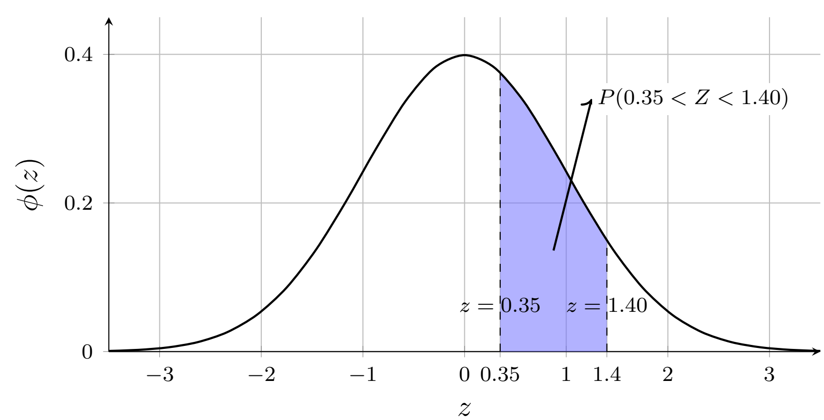
**Problem.** Let $Z \sim N(0, 1)$.

**Questions/Tasks.** Compute $P(0.35 < Z < 1.40)$.

## C5

The target central-band probability is

$$P(0.35 < Z < 1.40).$$



Using $P(a < Z < b) = \Phi(b) - \Phi(a)$, first read the cumulative areas:

$$P(Z < 1.40) = \Phi(1.40) = 0.9192$$

$$P(Z < 0.35) = \Phi(0.35) = 0.6368.$$

The central band is the difference of these cumulative areas:

$$P(0.35 < Z < 1.40) = \Phi(1.40) - \Phi(0.35) = 0.9192 - 0.6368 = 0.2824.$$

**Interpretation.** About 28.2% of values are between 0.35 and 1.40.

# 5 Problem 5: Central Band with a Negative and a Positive Bound

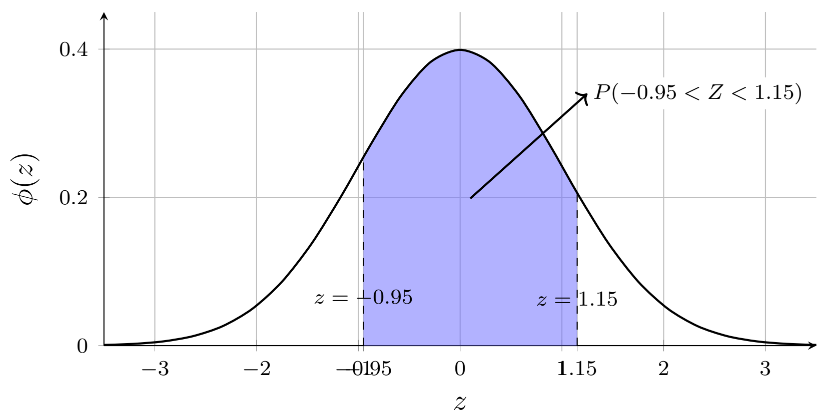
**Problem.** Let $Z \sim N(0, 1)$.

**Questions/Tasks.** Compute $P(-0.95 < Z < 1.15)$.

### C5

The target central-band probability is

$$P(-0.95 < Z < 1.15).$$




$$\Phi(-0.95) = 1 - \Phi(0.95) = 1 - 0.8289 = 0.1711.$$

$$\Phi(1.15) = 0.8749.$$

$$P(-0.95 < Z < 1.15) = 0.8749 - 0.1711 = 0.7038.$$

**Interpretation.** About 70.4% of values lie between -0.95 and 1.15.

# 6 Problem 6: Central Band with Two Negative Bounds

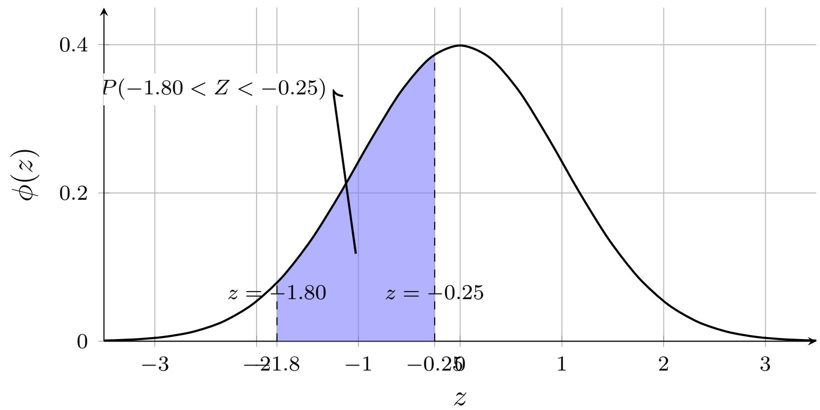
**Problem.** Let $Z \sim N(0, 1)$.

**Questions/Tasks.** Compute $P(-1.80 < Z < -0.25)$.

### C5

The target central-band probability is

$$P(-1.80 < Z < -0.25).$$




$$\Phi(-1.80) = 1 - \Phi(1.80) = 1 - 0.9641 = 0.0359$$

$$\Phi(-0.25) = 1 - \Phi(0.25) = 1 - 0.5987 = 0.4013.$$

$$P(-1.80 < Z < -0.25) = 0.4013 - 0.0359 = 0.3654.$$

**Interpretation.** About 36.5% of values are between -1.80 and -0.25.

## 7 Problem 7: Two Tails with Positive Bounds

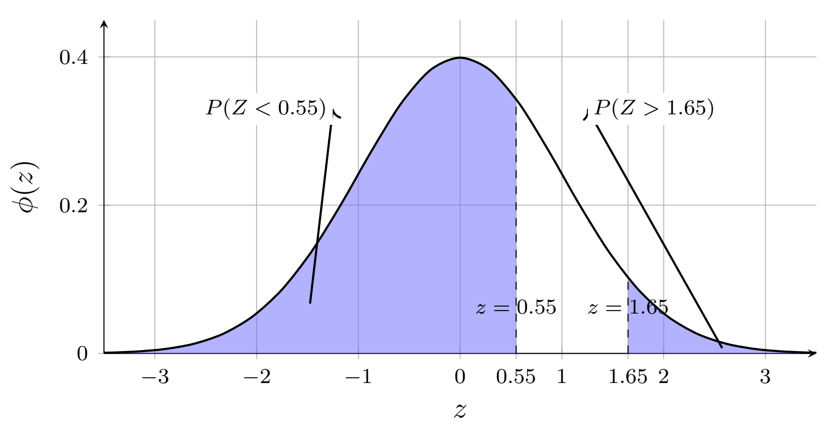
**Problem.** Let $Z \sim N(0, 1)$.

**Questions/Tasks.** Find $P(Z < 0.55 \text{ or } Z > 1.65)$.

### C5

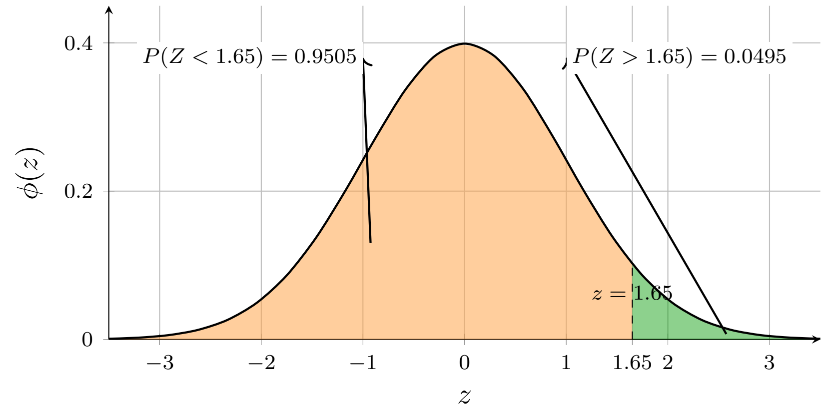
The target union probability is

$$P(Z < 0.55 \text{ or } Z > 1.65).$$




These two events are disjoint, so we add probabilities. Note that $Z < 0.55$ is a left region with a *positive* bound, not a negative tail.

$$P(Z < 0.55) = \Phi(0.55) = 0.7088.$$




$$P(Z > 1.65) = 1 - \Phi(1.65) = 1 - 0.9505 = 0.0495.$$

$$P(Z < 0.55 \text{ or } Z > 1.65) = 0.7088 + 0.0495 = 0.7583.$$

**Interpretation.** Most values satisfy this union because all values below 0.55 are included plus a small far-right tail.

# 8 Problem 8: Two Tails with One Negative and One Positive Bound

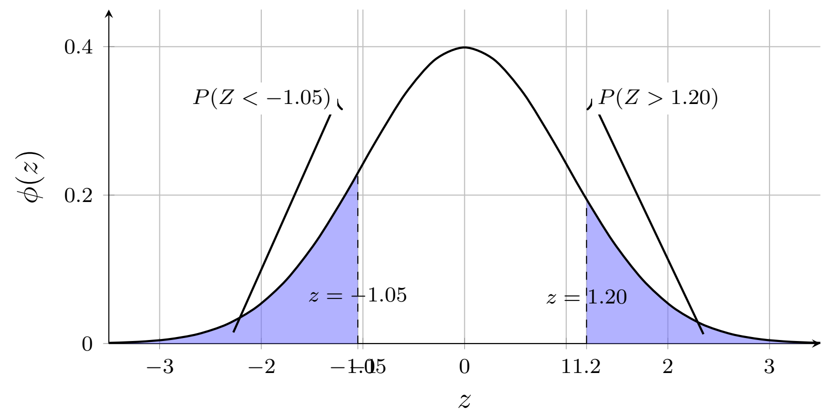
**Problem.** Let $Z \sim N(0, 1)$.

**Questions/Tasks.** Find $P(Z < -1.05 \text{ or } Z > 1.20)$.

### C5

The target union probability is

$$P(Z < -1.05 \text{ or } Z > 1.20).$$




The events are disjoint, so add them.

$$P(Z < -1.05) = P(Z > 1.05)$$

$$P(Z < -1.05) = 1 - \Phi(1.05) = 1 - 0.8531 = 0.1469.$$

$$P(Z > 1.20) = 1 - \Phi(1.20) = 1 - 0.8849 = 0.1151.$$

$$P(Z < -1.05 \text{ or } Z > 1.20) = 0.1469 + 0.1151 = 0.2620.$$

**Interpretation.** About 26.2% of values are in these two outer regions.

# 9 Problem 9: Two Tails with Two Negative Bounds

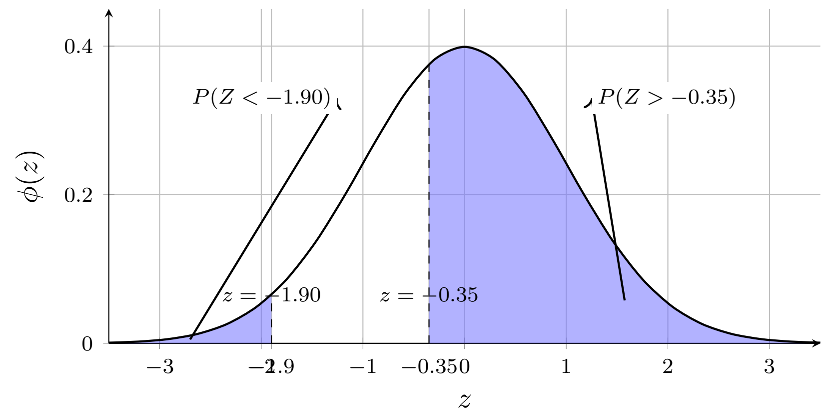
**Problem.** Let $Z \sim N(0,1)$.

**Questions/Tasks.** Find $P(Z < -1.90 \text{ or } Z > -0.35)$.

### C5

The target union probability is

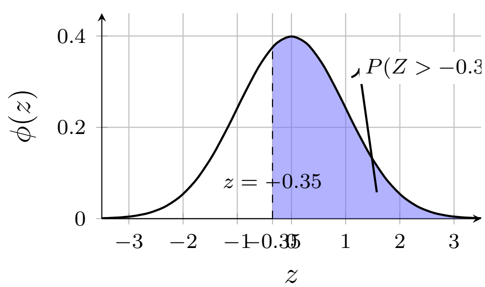
$$P(Z < -1.90 \text{ or } Z > -0.35).$$

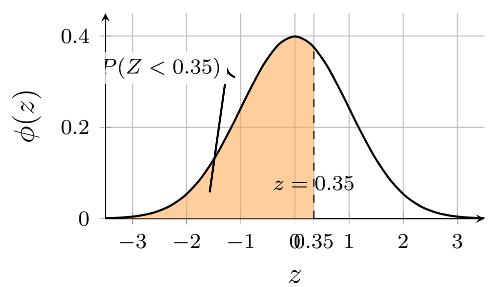



The two events are disjoint because one is far left of $-1.90$ and the other is to the right of $-0.35$.

$$P(Z < -1.90) = 1 - \Phi(1.90) = 1 - 0.9713 = 0.0287.$$

For the right-tail bound at a negative value:




$$P(Z > -0.35) = P(Z < 0.35) = \Phi(0.35) = 0.6368.$$

So

$$P(Z < -1.90 \text{ or } Z > -0.35) = 0.0287 + 0.6368 = 0.6655.$$

**Interpretation.** This probability is large because all values above -0.35 are included, plus a small far-left tail.